# Full Stack Assessment Case Study (AI Assisted)

**Project:** AI Interview Platform
**Assessment date:** 27 October 2026
**Review type:** Runtime check, PRD check, and engineering quality review
**Product references:** PRD 01: First Principles (AI Interview Behavior) and PRD 02: Real Simulation (End-to-End Interview)
**Assignment reference:** The *Fullstack Product Engineer* case-study brief I was given

## Executive summary

The API responds to health checks and the web app builds. A handful of assessor screens and API workflows also passed earlier checks using test data. That shows the services can run, but it does not show the product can complete the interview described in the PRDs.

The clearest bug a user would hit is in the candidate hardware check. When a candidate clicks Retry, the checks get stuck and Start Interview stays disabled. I reproduced this. The internet check also leans on outside test hosts, and those failed in this environment. Reading the source, I spotted two more concerns. The microphone may not get cleaned up, and an old WebSocket disconnect callback may fire late. I haven't seen either one happen.

The biggest gap is the AI interview itself. I never ran a real Gemini interview. So I can't say that follow-up probing, coverage tracking, the portfolio, fit/gap results, export, or reconnect work. The repo also has no test suite anyone can rely on. Rails reports zero tests, and RSpec crashes before it finds any.

**Overall conclusion:** The runtime and build checks look good. But end-to-end readiness isn't shown. The next milestone should be a repeatable test journey with made-up data, from assessor setup all the way to interview results. It should run on tests that don't need paid AI access, a real microphone, or outside network probes.

## What this assessment covers

This report pulls together the runtime checks and screenshots I already had, the app screens and API routes that exist, both PRDs, and the case-study brief. It's an engineering quality review. It isn't a security certification, a legal opinion, or the final PDF for the hiring submission.

### User and product context

The product is for assessors and recruiters who need solid evidence about a candidate's skills. It's also for hiring managers who have to decide based on that evidence. Candidates feel the effects too, even though they often don't choose the tool. A wrong or weakly backed rating can cost someone a real opportunity.

So a useful result has to be more than a level from an AI. It should point back to what the candidate actually said. A human reviewer should be able to follow it. And it should be upfront about what it doesn't know.

Candidate audio, transcripts, results, and exports need care too. That means a clear purpose for the data, access controls, retention rules, and honest notice to the candidate. The brief says to consider Indonesia's Personal Data Protection Law (UU PDP). The suggestions below are product and engineering safeguards. They aren't legal advice, and they don't say whether the product complies.

### Method and limits

- I read both PRDs and the brief. Then I compared their workflows with the routes in the app and with the smoke-test results I already had.
- I checked the local API health endpoints and ran the web production build.
- I reused earlier smoke checks that ran on test data. They covered sign-in, skill lookup, assessment and session routes, and vacancy create, read, update, and delete. I removed the temporary test records afterward.
- I reviewed the candidate preflight and WebSocket findings by reading the source code.
- I used existing screenshots as evidence for the candidate preflight and assessor setup screens.
- I did a limited security review. It found no confirmed vulnerability. But I didn't try real exploit payloads, and I didn't test every authorization boundary or API input.
- My last browser automation run used a made-up candidate response. The test environment blocked microphone permission, and its internet check didn't finish in time. That's a limit of the test setup. It doesn't prove every candidate device will fail.
- I never printed a Gemini API key or sent one to Google. Earlier Gemini checks used local stubs only. So I haven't confirmed real provider access, free-tier quota, billing status, or live audio behavior.

## Runtime check

| Area | Result | What it shows, and what it doesn't |
|---|---|---|
| Rails API | **Responding** | `/health` and `/api/v1/health` returned HTTP 200. That only shows the API is alive. It doesn't prove every business workflow works. |
| PostgreSQL, Redis, Sidekiq | **Confirmed connected earlier** | In the earlier setup pass, I confirmed a local database query, a Redis ping, and one Sidekiq process. |
| Web application | **Build passed** | `npm run build` in `web/` passed the TypeScript check and the Vite production build. |
| Assessor screens | **Some screens passed** | Login, the assessment list, the new-assessment form, and the skill picker loaded in earlier checks with test data. |
| Candidate preflight | **Not reliable in this run** | Outside test requests failed in the earlier browser run, and Retry left the checks waiting. |
| Full live interview | **Not verified** | I didn't run a real Gemini Live session, and I didn't finish an interview with a microphone. |
| Automated tests | **Can't catch regressions** | Rails ran zero tests. RSpec failed while loading `spec_helper`. The web package has no test script. |

### Commands and what happened

- `GET /health`: **HTTP 200**, `{"status":"ok"}`.
- `GET /api/v1/health`: **HTTP 200**, `{"status":"ok"}`.
- `npm run build` from `web/`: **passed**, including the TypeScript check.
- `bundle exec rails test` from `api/`: **0 runs, 0 assertions, 0 failures, 0 errors**.
- `bundle exec rspec` from `api/`: **failed before running any examples** with `LoadError: cannot load such file -- spec_helper`.
- Earlier smoke tests: test admin login, skill lookup (22 seeded skills), assessment and session API checks, and vacancy create, read, update, and delete. All **passed on the paths I tried**. I removed the temporary test data afterward.

## PRD alignment: today's product against the PRDs

I kept these statuses cautious on purpose. "Code is there" means a route, screen, or service exists. It doesn't mean the behavior passed a test.

### PRD 01: First Principles (AI Interview Behavior)

| What the PRD expects | What I found | Status |
|---|---|---|
| The interviewer reacts to what the candidate actually said and asks relevant follow-ups. It doesn't just read a fixed list of questions. | The PRD describes this and gives examples. I had no live interview to check whether the model does it. | **Unverified** |
| It probes vague claims, pushes back on strong claims only as hard as they call for, and looks into useful skills that come up off the agenda. | These are core PRD behaviors. I had no real-time model session or transcript to look at. | **Unverified** |
| It keeps questions short, moves between topics naturally, skips empty praise, and wraps up at the right time. | The PRD includes prompt guidance. I didn't test what the model actually says. | **Unverified** |
| It tracks coverage states and requires enough probing before it marks a skill covered. | Coverage routes and services exist. But I didn't run the live sequence or the state changes. | **Code is there; behavior unverified** |
| It builds a portfolio from the candidate's evidence, rubric anchors, confidence, and coverage. | Portfolio routes and generation code exist. There was no finished interview or generated portfolio, so I couldn't check the evidence quality or the ratings. | **Code is there; output unverified** |
| It avoids the failure modes the PRD lists, like skipping follow-ups or ending before skills are covered. | I ran no realistic interview, and there's no automated way to check the model's behavior. | **Unverified** |

**PRD 01 conclusion:** The PRD describes the behavior it wants in detail. But I can't confirm the running AI does any of it. The earlier checks used stubs in place of Gemini. They only show that the app handles requests and responses locally. They say nothing about interview quality, model availability, or whether the scoring is valid.

### PRD 02: Real Simulation (End-to-End Interview)

| Stage of the simulation | What I found | Status |
|---|---|---|
| The assessor signs in, sets the role, time, and skills, and creates an invite. | Login, the assessment list, the form, and the skill picker loaded with test data. I tried some setup and API actions. But I didn't complete the full path from the UI to an invite, and I didn't run the Gemini prompt generation end to end. | **Partially verified** |
| The candidate opens the invite and sees the right role and session details. | The candidate-info lookup passed in an earlier local API smoke test. The final browser page used a made-up candidate response, not a real saved invitation. | **Partially verified** |
| The candidate allows the microphone and starts the interview. | The browser automation environment blocked microphone access, so I never got to the start of the interview. | **Not verified** |
| Gemini Live opens and the session goes active. | I never started a real provider session. | **Not verified** |
| Interview turns are saved, and follow-ups and coverage update as the interview goes on. | There was no live interview, transcript, or coverage timeline. | **Not verified** |
| The session ends naturally or when the candidate stops it, and that kicks off portfolio generation. | I had no finished live session. | **Not verified** |
| The assessor reviews skill levels and the evidence behind them. | There was no generated portfolio to review. | **Not verified** |
| Fit/gap compares the portfolio with a vacancy and gives a useful write-up. | Vacancy create, read, update, and delete passed in smoke testing, and fit/gap routes exist. But no real portfolio-to-vacancy result was generated. | **Code is there; end-to-end behavior unverified** |
| Export produces a usable result document. | The export route exists. There was no generated result to export or look at. | **Code is there; output unverified** |
| A Gemini reconnect picks up the conversation without losing state. | I didn't run a controlled Live API reconnect test. A separate concern about a browser-disconnect race, found in the source, is listed below. | **Not verified** |

**PRD 02 conclusion:** The assessor setup and a few API routes are the best-verified parts. The journey from candidate to results is the core promise of the product. And it's still unproven.

## Findings and severity

I set severity by how badly a user is hurt and how likely it is in the workflow I tested. I confirmed no P0. If a finding says "source-reviewed" or "unverified," I'm not claiming I saw it happen.

### P1: Retry can leave a candidate unable to start

**Service:** Web **Type:** Code bug; reproduced **Impact:** A candidate hits a failure they could recover from. Then Retry leaves the checks stuck on "Checking" or "Waiting," and Start Interview stays disabled.

So here's what happened. The browser showed an Internet failure. After Retry, the OS/browser check stayed on "Checking..." and the Internet check stayed on "Waiting." They never settled. In `HardwareCheck.tsx`, Retry resets the statuses. But at least one check starts from an effect that only runs on mount. Resetting the status doesn't run that effect again. So that check never restarts.

**Recommended fix:** Treat each preflight attempt as its own thing. On Retry, restart every check. Cancel or ignore results from the previous attempt. Then add a browser test that fails before the fix and passes after it.

## P1: Outside network probes can block a candidate

**Service:** Web **Type:** Blocker in this environment, plus a weak spot in the product. I saw it fail here. **Impact:** If the third-party CDN or echo hosts the code uses can't be reached, a candidate can be blocked even when the app's own API is fine. Picture a candidate on a good connection whose network blocks those hosts. They could still get stuck.

`internetSpeedTest.ts` hard-codes third-party download sources and uses default upload hosts. In the earlier browser run, those requests failed, and the check returned failing values. That shows the app depends on those hosts, and that they failed here. It doesn't show that every production network has the problem.

**Recommended fix:** Use a probe the team owns, or one that can be configured. Decide what happens when a measurement isn't available. Tell an unreachable test host apart from a connection that's truly too slow. And don't let a nice-to-have outside measurement be the only gate to starting.

## P2: Microphone and audio-monitoring cleanup needs checking

**Service:** Web **Type:** Concern from reading the source (resources and privacy); not reproduced **Impact:** A mic-only stream may stay open after a retry or after the component goes away. And the audio-level animation loop may keep running after its component is gone.

The cleanup code in `HardwareCheck.tsx` stops the tracks held in the video stream state. But it doesn't clearly keep the mic-only stream around so it can be stopped later. The animation loop calls `requestAnimationFrame` and has no way to cancel it. The browser couldn't grant microphone permission, so I never saw real device cleanup.

**Recommended fix:** Keep a handle on every media track the code opens. Stop each one on retry, on completion, and on unmount. Cancel the animation frames and close the audio contexts. Test it with a fake media stream and a controlled animation-frame scheduler.

## P2: An old WebSocket disconnect action may hit a reconnected session

**Service:** API **Type:** Possible race from reading the source; not reproduced **Impact:** Say a candidate reconnects during the grace period. A delayed action from the earlier disconnect could still fire and act on a session that's live again.

The disconnect flow I reviewed is in `audio_websocket_middleware.rb`. I didn't run a controlled timer and reconnect test. So this is a risk to check. It's not a confirmed bug where a session gets ended by mistake.

**Recommended fix:** Cancel the old timer, or guard it with a token that marks which disconnect it belongs to. Before ending the session, check two things: the browser is still disconnected, and the callback belongs to the current disconnect. Then test both cases with a fake clock. One is a reconnect before the timer runs out. The other is a timer that runs out with no reconnect.

### P2: The automated tests can’t protect the workflow right now

**Services:** API and web **Type:** No working safety net **Impact:** CI has no reliable check that would catch the Retry bug, or the cleanup and reconnect risks, if they came back.

Rails finds zero tests. RSpec fails when it loads `spec_helper`. The web package has build and dev scripts but no test command. A passing build doesn’t replace workflow tests.

**Recommended fix:** Get working test runners for both the API and the web app. Make CI fail when it finds zero tests. Cover the candidate flow, session handling, and the failure paths with fixed, repeatable test data.

### P3: Candidate transparency and assessor review need clear product requirements

**Services:** API and web **Type:** Gap in the product and privacy requirements; not a bug I proved in the running workflow **Impact:** If nobody decides how candidates get notice, how evidence is reviewed, how mistakes are corrected, and how long data is kept, teams may treat an AI rating as a full hiring decision. Neither the candidate nor the assessor would have enough context.

The brief asks the team to think about candidates who never chose this product, and to consider UU PDP. The PRDs focus on how the interview behaves and what it produces. I didn’t find a tested, end-to-end experience for candidate transparency, rating review, correction requests, or data retention. That’s a gap in the spec and in testing. I’m not saying the app breaks the law.

**Recommended next step:** Before any real candidate data goes in, sit down with product, privacy, and legal people. Decide what the candidate is told. Decide who can see the data and how long it’s kept. Decide who is accountable on the assessor side. And decide how a rating gets reviewed or corrected.

### Security review

The limited security review found **no confirmed vulnerability**. It wasn’t a full audit yet, some areas still need repeatable tests such as authentication and authorization on the result and export routes, invite-token handling, input validation, dependency risk, and how candidate data is stored, logged, and kept. These are the recommended places to test and aren’t reported vulnerabilities.

## Case-study brief alignment

The brief asks for a product-engineering submission, not just an app that runs. Here’s what this report covers and what it doesn’t.

| Brief step or deliverable | Status in this assessment |
| --- | --- |
| Run and explore both services | **Done, as far as recorded above.** I checked API health, the connections to PostgreSQL, Redis, and Sidekiq, the web build, some assessor screens, and the API smoke checks. |
| Understand users and candidate impact | **Partly covered.** The report names assessors, recruiters, hiring managers, and candidates, and says why evidence quality and privacy matter. I didn’t do independent research on hiring in Indonesia. And there was no legal review under UU PDP. |

| Brief step or deliverable | Status in this assessment |
| --- | --- |
| Rank gaps from P0 to P3 and separate missing definition from broken behavior | **Covered for the findings here.** Retry is a bug I reproduced. The reliance on outside probes is a problem I saw in this environment, and also a design gap. Media cleanup and reconnect are concerns from reading the source. The test setup is missing. I confirmed no P0. |
| Propose a full-stack revamp with acceptance criteria and trade-offs | **Covered as a proposal below.** I didn't build any fix or new feature as part of this report update. |
| Prove tests fail before a fix and pass afterward | **Not done.** I didn't run an experiment where I plant a fault on purpose. |
| Show migration safety, failure handling, and automated implementation tests | **Not shown by this review.** The API has no real tests that could back those claims up. |
| Document an AI-generated suggestion that was risky or wrong, and how it was corrected | **Not fully met.** I checked earlier model defaults with local stubs. But that isn't the AI-verification moment the brief asks for, and I'm not claiming it is. |
| Provide a pull-request link and a 3 to 5 minute walkthrough video | **Not included.** This report doesn't create a PR or a video, and it doesn't claim to. |
| Provide visual evidence | **Partly covered.** The screenshots show the candidate preflight failure and retry, and the assessor setup. They don't show a complete or redesigned interview and results flow. |
| Submit a single PDF | **Not produced.** What I was asked for here is the Markdown report. Turn it into a PDF only if the final submission portal needs one. |

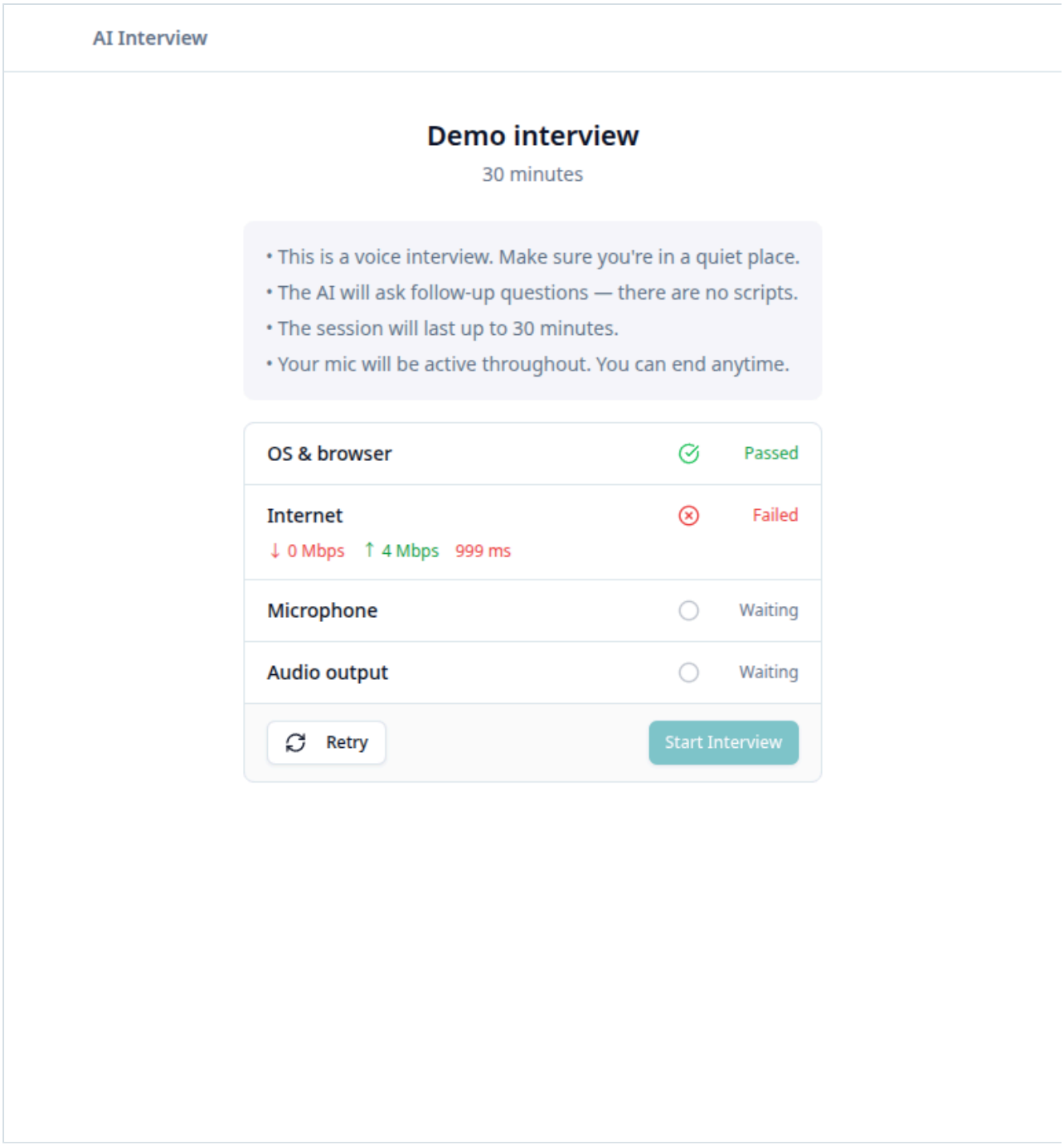
### What a technical lead should know

- **Provider behavior is unknown:** I sent no real Gemini request. Model and API compatibility, free-tier quota, rate limits, and live audio behavior all still need a smoke test that someone approves separately.
- **The core journey has no test protection:** Rails finds no tests, and RSpec can't load its helper. A passing build or a healthy health check doesn't show the product is right.
- **Candidate devices are barely tested:** The browser automation environment denied microphone permission, and outside speed-test services failed in the earlier run. The team needs a controlled device and media test, plus a clear fallback rule.
- **Someone has to own privacy:** Before real candidate data is used in production, the team needs a written decision on notice, purpose, access, and retention and deletion. People qualified to judge UU PDP should review it.

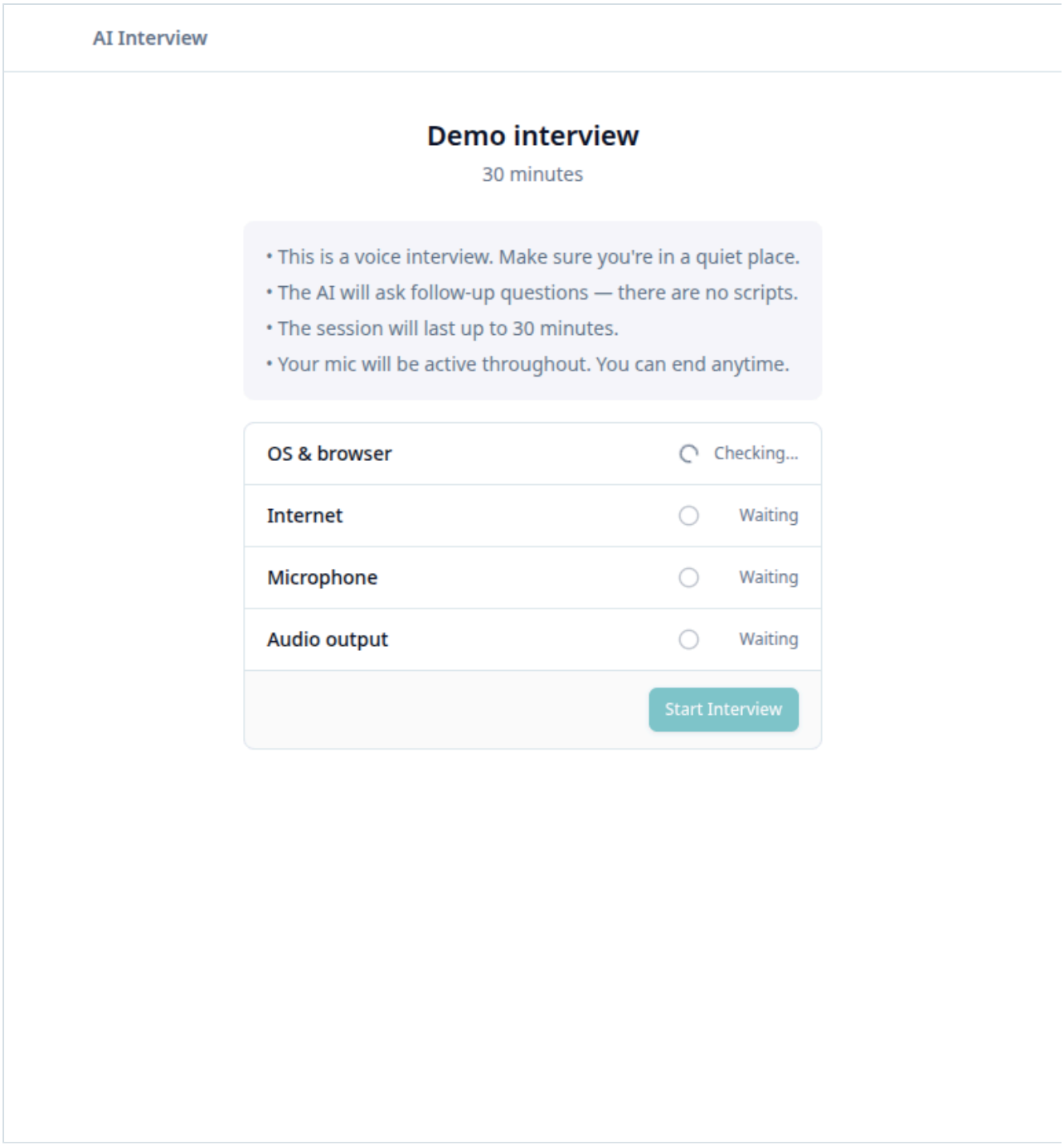
## Evidence

The assessor screenshots show test assessment data. They prove what was on the screen. They don't prove the full end-to-end workflow passed.

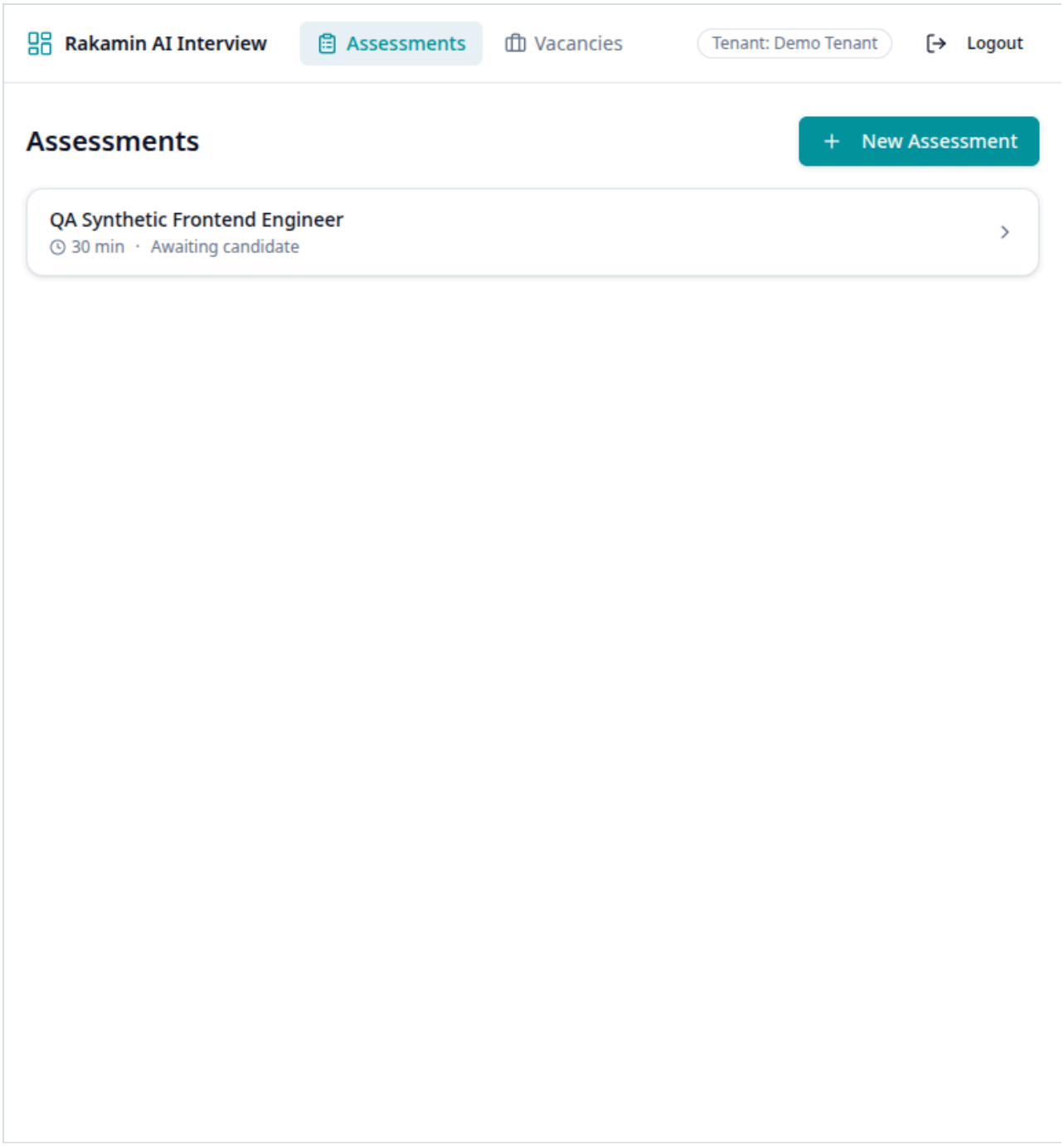
1. Candidate internet check failed



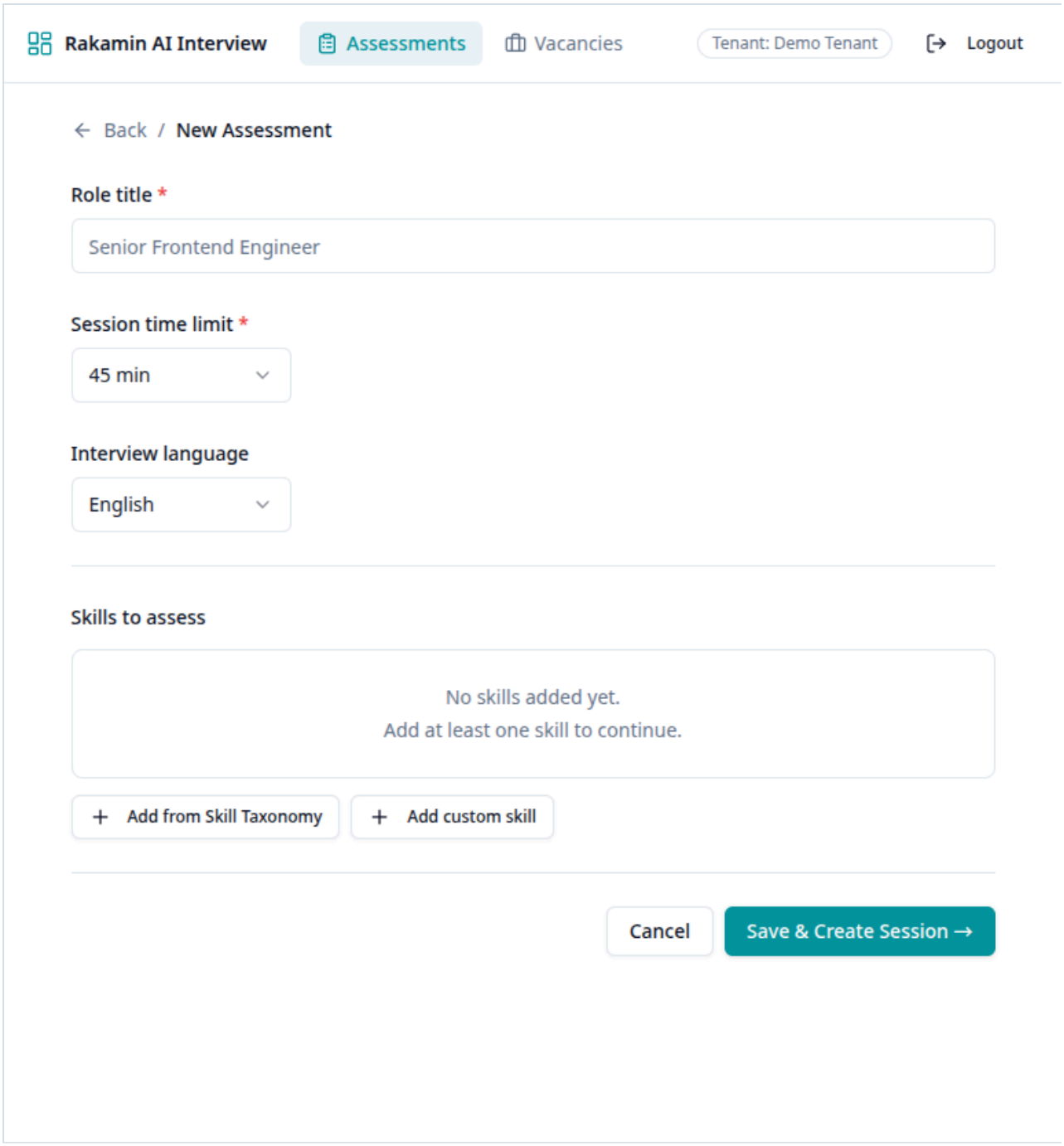

2. Retry leaves checks waiting



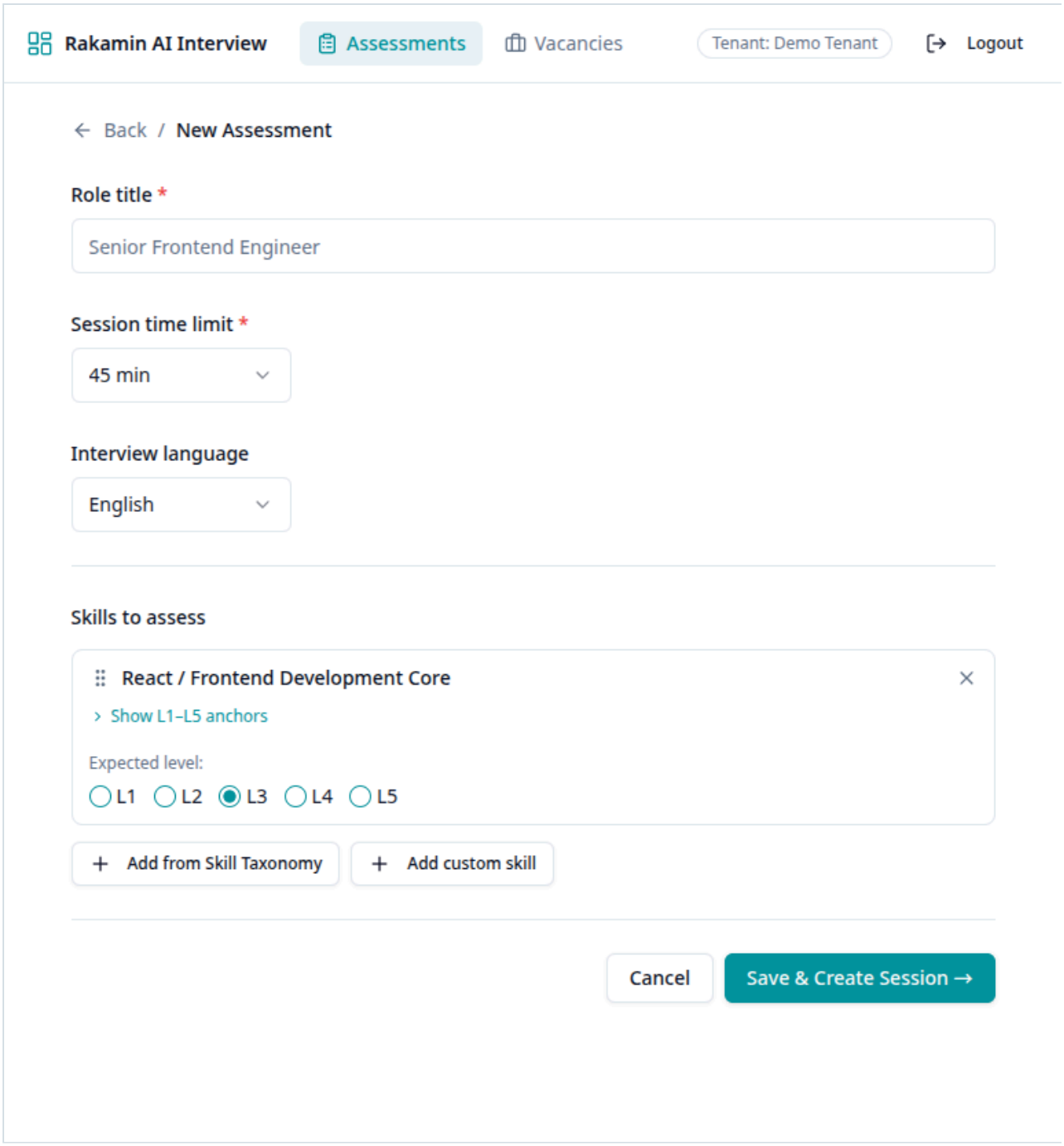

3. Assessor assessment list




4. New assessment form




5. Assessment skill selection




The microphone cleanup and WebSocket reconnect concerns need controlled automated tests. A screenshot can't prove timing problems or leaked resources.

## Proposed revamp: reliable interview results backed by evidence

Here's the thing. The best first step isn't another AI feature. Fix the candidate journey so a candidate can recover when something goes wrong. Then make sure every result an assessor sees can be traced to evidence, can be reviewed, and says what it doesn't know.

## Options and trade-offs

| Option | What it delivers | Cost and upkeep | Risks and trade-offs |
|---|---|---|---|
| **A. Patch only the browser checks** | Quick fixes for Retry and mic cleanup. Very little backend change. | Cheapest to start. Easy to maintain. | Leaves the outside probe dependence and the session-handling races alone. Fixes the symptoms without checking the whole journey. |
| **B. Build a reliable preflight and session slice** | A clear state for each preflight attempt, controlled probe behavior, media cleanup, session state that survives a reconnect, and tests across web and API. | Medium effort across both services. Leaves clean places to build on later. | Needs careful coordination between client and server state, and stable fakes for outside services. A bit more design up front. |
| **C. Build a bigger AI scoring and assessor decision platform first** | Adds rubric versions, portfolio review, overrides, and richer fit/gap workflows. | Most expensive. Touches the data model, prompts, UI, and permissions. | Could magnify interview and evidence problems nobody has checked yet. Costly to change if the live journey underneath turns out to be unreliable. |

**Recommendation:** Go with Option B first. It fixes a problem candidates actually hit. It makes the data that later AI analysis relies on more reliable. And it gives you pieces you can test. Once a repeatable test interview passes, follow up with the evidence-based review features from Option C.

## Acceptance criteria for the first revamp

1. **Retry:** Every check starts fresh on each attempt. Late responses from an earlier attempt can't overwrite the newest result.
2. **Network measurement:** If a probe is blocked or times out, the app says "could not measure." It doesn't quietly treat that as a measured zero. The product policy says whether that case blocks the interview from starting.
3. **Microphone:** Permission denied, no device, unsupported browser, and device already in use each show a different message. Each message tells the candidate what to do next.
4. **Cleanup:** On retry, on completion, on navigation, and on unmount, every audio track is stopped, every animation frame is cancelled, and every audio context is closed.
5. **Session start:** Sending the start request twice doesn't create two active sessions or two jobs. For an invitation that already started or has expired, the API returns a clear state.
6. **Reconnect:** If the candidate reconnects before the grace period ends, the session stays active. If the grace period ends with no reconnect, the session ends once. An old timer can't end a newer connection.
7. **Coverage:** A skill can't be marked covered unless it hit the PRD's minimum probe count and has evidence you could defend. If the analyzer call fails, the last known coverage map stays. It doesn't vanish without a word.
8. **Portfolio:** Every rating has evidence that came from the candidate, plus a confidence level. There's also a clear "insufficient evidence" state. A missing rating is never turned into a zero or into a nice-looking default.
9. **Human review:** Assessors can look at the evidence and correct a rating, and they write down why. The AI score is never shown as a final hiring decision nobody can question.
10. **Failure handling:** A Gemini timeout, a quota denial, malformed JSON, a partial transcript, or a background job delivered twice all leave a visible state that can be recovered. None of them produce an empty report

that looks like a success.

11. **Long content and odd inputs:** Long transcripts, empty strings, missing skill anchors, unexpected rating values, and Unicode or Indonesian text are all checked and shown properly. Nothing gets cut off in a way that changes the meaning, and nothing overflows the UI.
12. **Data protection:** Logs and test data hold no real candidate data and no secrets. Only authorized people can reach transcripts, portfolios, and exports. Retention and deletion rules are written down before production use.

## Future enhancements

These ideas come from the PRDs and the case-study brief. They're proposals. They aren't claims about what the product does today, and they aren't legal requirements.

### 1. Candidate trust and accessibility

- Add a clear notice before the interview. Say what gets recorded, why, who can see the results, and how long the data is kept.
- Give plain microphone and network diagnostics, with tips for fixing problems. Add another way forward for when device permissions or network measurement aren't available.
- Offer Indonesian and English for both the UI and the interview. Make keyboard and screen-reader use work. Make layouts fit different screens. And make reasonable accommodations for candidates with different devices or communication needs.
- Set up a proper way for candidates to ask questions, ask for corrections, or make data-rights requests. Have legal and operations review it before launch.

### 2. Evidence-based assessor review

- Show each skill rating with its rubric anchor, quotes from the candidate, confidence, probe count, and any evidence that's missing.
- Let an assessor accept or change an AI suggestion, with a short reason and a history of who changed what.
- Keep three things clearly apart: what was observed, what the model read into it, and what a human decided.
- Label skills found along the way as exploratory. Don't treat them like fully assessed skills.
- Allow a "not enough evidence" result and suggest a follow-up. Don't force a number.

### 3. Fair and consistent assessment

- Version the rubrics and the generated prompts. That way old results still make sense after a rubric changes.
- Add calibration tools for assessors. Use shared made-up examples, and let assessors compare the reasons behind their ratings.
- Watch for results that differ in a consistent way across language, role, or candidate groups. Combine the data in a way that protects privacy, and have qualified people review it.
- Don't guess protected or personal traits from a voice, an accent, appearance, or off-topic conversation.
- Treat automated scores as help for a decision, not the decision. A real person stays accountable.

### 4. Privacy, security, and operations

- List what data exists, and set its purpose, retention, deletion, and access rules for audio, transcripts, portfolios, and exports. Bring in someone who knows UU PDP.
- Keep a record of who accesses candidate records and exports. Strip secrets and extra personal data out of the logs.
- Use role-based access, and check authorization explicitly on the portfolio, transcript, fit/gap, override, and export endpoints.
- Add the team's own health and readiness checks. Make queues and jobs visible. Sort provider errors into clear categories. Set up alerts for operations.
- Let background jobs retry and recover in a controlled way, without duplicate portfolios or half-finished writes.

### 5. Hiring-team workflow

- Support reviewer comments and structured follow-up questions that are tied to evidence. A single score isn't enough.
- Compare candidates against a role rubric. Don't rank them on signals nobody validated, or signals that don't matter for the job.
- Let teams share assessment templates. Each template should have an owner, a version history, and clear permissions.
- Make exports accessible, tied to a rubric version, and clear about which conclusions came from the AI and which came from a person.

## Phased delivery plan

### Phase 1: Make the tests something you can trust

- Fix the API test setup, or switch to a runner that's clearly supported. Add a check that fails when no tests are found.
- Add a web test runner and repeatable browser tests for the candidate preflight.
- Keep test data separate from the development seed data. Keep normal CI free of live Gemini, real microphones, email, and outside CDNs.

**Exit check:** CI runs real API and web tests, and it fails when it can't find them.

### Phase 2: Steady candidate entry and session handling

- Fix how Retry attempts run, how the network probe behaves, media cleanup, and how old reconnect timers get cancelled.
- Add WebSocket tests with a fake clock. Add fake media streams, timeout tests, and permission-denied tests.
- Check how the API handles an expired invite, a duplicate start, a repeated end, and an interrupted session.

**Exit check:** A test candidate can recover from a preflight failure, start once, reconnect safely, and leave no microphone resources running after exit.

### Phase 3: Check interview behavior without paid AI

- Use a fake Gemini adapter and canned transcripts. Test the prompt setup, transcript saving, coverage changes, minimum probing, and the conditions that end a session.
- Add scenario checks for vague answers, strong answers, off-agenda finds, short answers, silence, and model errors.
- Confirm the results never give a skill level the evidence doesn't support.

**Exit check:** The PRD's interview rules are covered by repeatable automated tests before anyone judges real model behavior.

### Phase 4: Build the evidence-based portfolio and human review

- Add versioning for rubrics and prompts. Add evidence and confidence fields to the portfolio.
- Add assessor review, rating overrides with a reason, and handling for "insufficient evidence."
- Test portfolio generation failures, duplicate jobs, partial outputs, and long transcripts.

**Exit check:** Every rating on screen traces back to evidence and a rubric version. An assessor can correct it without losing the history.

### Phase 5: Run the full PRD simulation and a controlled provider check

- Run assessor setup, invite, candidate preflight, live interview, coverage, portfolio generation, fit/gap, and export, all with test data.
- Add provider smoke checks that people choose to run. They should hide secrets, respect the quota, and stay separate from CI.
- Use a short test session to check current model and API compatibility, timeout and quota behavior, and reconnecting and resuming.
- Record screenshots and a walkthrough of a success path and of a real failure being recovered.

**Exit check:** The whole PRD scenario passes with test data. Any claim about the live provider is backed by a separate, safe smoke test that was recorded.

### Phase 6: Get ready for real candidates

- Review privacy, retention, access, accommodations, and what candidates are told with the right legal and domain experts.
- Run systematic authorization and input validation tests across the candidate and assessor APIs.
- Set up monitoring, incident response, backup and restore, migration rollback, and release readiness checks.

**Exit check:** Product, engineering, and privacy people agree on the operating controls before any real candidate data goes in.

## Final assessment

The project runs well enough to build and to answer health checks. I also tried a few assessor and API paths. But I can’t say yet that the product works end to end against the PRDs. The candidate Retry bug is reproduced. Several AI, reconnect, result, and export behaviors are still unverified. And there’s no automated suite to close those gaps of the feature.

The right next step is a small testable slice that makes the full stack more reliable. First, make candidate entry recoverable. Next, make session handling safe. Then prove the full test interview and the evidence-based review flow. That gives assessors a tool they can depend on. And it gives candidates a fairer experience they can understand.

# Future enhancement UI concepts

A visual companion to the five recommendations in the assessment. Fifteen proposed screens, three per theme.




## Design index

A cohesive direction based on the existing teal-and-white interface.

15 screen concepts

### 01 · Candidate trust & accessibility

3 screens

01 A clear, plain-language interview notice

02 Guided device check with a safe fallback

03 Candidate privacy and support center

### 02 · Evidence-based assessor review

3 screens

01 Skill review anchored in candidate evidence

02 Coverage map and evidence-gap follow-up

03 Human decision history and review provenance

### 03 · Fair & consistent assessment

3 screens

01 Rubric versioning with a transparent change log

02 Assessor calibration with shared practice cases

03 Aggregate fairness monitoring with privacy safeguards

### 04 · Privacy, security & operations

3 screens

01 Data inventory and retention policy controls

02 Role-based access and auditable record access

03 Queue health and recoverable provider failures

### 05 · Hiring-team workflow

3 screens

01 Candidate comparison against role requirements

02 Shared assessment templates with ownership and versions

03 Evidence-linked reviewer discussion and accessible export

**Concepts only:** These pages illustrate future possibilities, not existing functionality. Example data, policy values, and states are illustrative; confirm product, technical, security, accessibility, and legal requirements before implementation.

# A clear, plain-language interview notice



Rakamin AI Interview

English ▾ | Bahasa Indonesia

Invitation / Before you begin

## Before your interview

Step 1 of 3

A few minutes to understand what to expect and how your information is used.

**Your choice matters.** Please review this notice before continuing. You can contact the hiring team if you have questions or need an accommodation.

### What may be collected

- Microphone audio during the interview — Only in session
- Transcript and interview answers — For review
- Device readiness results — Troubleshooting

### Why, who, and how long

**Purpose:** assess skills for the role you applied to.

**Access:** the authorized hiring team for this role.

**Retention:** kept under the employer's published retention schedule.

Read the full privacy notice ↗

☐ I understand how this interview works and where to ask questions.

☐ I need an accommodation or another format.

Ask the hiring team

Continue to device check →

**What it shows:** Explains what the interview collects, why it is used, who can see it, and where to ask questions before a candidate proceeds. **Why it matters:** Candidates should be able to make an informed choice and raise accommodation needs before recording or device checks begin.

# Guided device check with a safe fallback

Candidate entry · Distinguishes measured results from unavailable checks and gives next steps.

Rakamin AI Interview

English ▾ | Bahasa Indonesia

## Check your setup

Step 2 of 3

You can retry a check or ask for help. Results are saved for this attempt only.

### Microphone

Ready

Built-in microphone · Input level detected

Test microphone

### Browser & permissions

Supported

Allow microphone access when your interview begins. Camera is not required.

### Network check

Could not measure

The test service did not respond. This is not a measured speed result.

You can continue if your connection is stable. If the interview disconnects, use the reconnect link or contact support.

Try again

No score recorded

Keyboard tip: use Tab to move through controls; press Enter to activate a button. Screen-reader labels are available for each check.

Back

Continue to interview →

**What it shows:** Shows microphone, browser, and network outcomes separately, with actionable retry guidance and an honest “could not measure” state. **Why it matters:** Candidates need a recoverable path through device problems; unavailable network measurements must not be presented as a failed score.

# Candidate privacy and support center

Candidate self-service · A transparent path to questions, corrections, and data requests.

Rakamin AI Interview

English ▾ | Bahasa Indonesia

Candidate help / Privacy and support

## Your information and support

Secure help center

See how to reach the hiring team about your interview record.

### Interview record

| | |
|---|---|
| Audio recording | Not retained in this session |
| Transcript and assessment | Retention per employer policy |

Access is limited to the authorized team for this role. For the exact retention date, ask the employer.

### What would you like help with?

Ask a question | Request a correction | Data rights request

Requests are routed to the employer's designated contact. This screen does not promise a legal outcome.

### Your requests

**Transcript correction** — In review

Submitted 24 Oct · Ref. CS-1042

"Please review the spelling of my project name."

Need more help?

**people-team@example.org**

View request history

**What it shows:** Provides a candidate-facing place to understand the record, ask questions, request corrections, and follow a data request. **Why it matters:** A clear support route builds trust and gives candidates a practical way to challenge inaccuracies or seek help after an interview.

# Skill review anchored in candidate evidence



Rakamin AI Interview | Assessments | Vacancies | Review | Tenant: Demo Hiring Team | Help

Assessments / Senior Frontend Engineer / Session 08 · Portfolio review

## Review skill evidence

Candidate: Jordan Lee · Interview completed 26 Oct · Human review required

Open transcript ↗

### Skills in this assessment

- **React architecture** — Review
  2 of 3 probes · evidence available
- Testing strategy — Evidence complete
- Accessibility — 1 probe · gap

### React architecture

Suggestion, not decision

AI suggestion: Level 3 · Confidence: Medium · 2 probes

OBSERVED · 18:42

"I split the search state from the results list so typing doesn't trigger a full page refresh."

MODEL INTERPRETATION

Describes a separation of concerns and a user-facing performance trade-off. One edge case remains unexplored.

RUBRIC ANCHOR · LEVEL 3

Chooses a suitable structure and explains trade-offs; identifies common failure modes.

View rubric

### Assessor decision

Not yet reviewed

Choose a rating only if the evidence supports it.

Insufficient evidence | Level 2 | Accept Level 3 | Level 4

Reason for decision (required when changing the AI suggestion)

**What it shows:** Presents a skill's candidate quote, AI interpretation, rubric anchor, coverage, confidence, and separate assessor decision. **Why it matters:** Assessors can verify what was actually said, question model interpretation, and make an accountable decision instead of relying on an opaque score.

# Coverage map and evidence-gap follow-up



Rakamin AI Interview | Assessments | Vacancies | Review | Tenant: Demo Hiring Team | Help

Assessments / Session 08 / Coverage

## Evidence coverage

1 follow-up suggested

Coverage is a prompt for review, not a measure of candidate quality.

### Coverage by skill

| SKILL | PROBES | EVIDENCE | REVIEW STATE |
|---|---|---|---|
| React architecture | 2 / 3 | 2 quotes | One gap |
| Testing strategy | 3 / 3 | 3 quotes | Ready |
| Accessibility | 1 / 3 | 1 quote | Insufficient |

Open transcript at 18:42

Review evidence

Not enough evidence

### Accessibility

No rating is assigned. A follow-up could ask how the candidate validates keyboard navigation and screen-reader behavior.

> Candidate mentioned semantic HTML, but did not describe a validation approach.

Leave unrated

Add follow-up

**Fairness guardrail:** Don't infer ability from an unanswered prompt or a connectivity interruption.

**What it shows:** Maps probes and evidence by skill, highlights gaps, and offers a follow-up or an explicit insufficient-evidence outcome. **Why it matters:** Visible gaps discourage unsupported ratings and help teams distinguish missing evidence from a candidate's lack of ability.

# Human decision history and review provenance



Rakamin AI Interview | Assessments | Vacancies | Review | Tenant: Demo Hiring Team | Help

Assessments / Session 08 / Portfolio / React architecture

## Decision history

The AI suggestion remains visible beside the accountable human decision.

Back to portfolio

### Current reviewed result

Human-reviewed

**OBSERVED EVIDENCE**

Candidate described separating search state and results to avoid a full refresh.

Transcript quote · 18:42

**AI INTERPRETATION**

Level 3 suggestion, medium confidence. One edge case was not explored.

**HUMAN DECISION**

**Level 3 · Alex Morgan**

"The example meets the level-3 anchor; no evidence supports a higher level."

Edit decision

### Change history

- **27 Oct, 10:14 · Alex Morgan**
  Accepted Level 3 after reviewing quote and rubric anchor.
- **26 Oct, 17:52 · Analysis completed**
  AI suggested Level 3 · Medium confidence · rubric v2.1
- **26 Oct, 17:48 · Interview completed**
  2 probes recorded · 2 evidence excerpts linked

Edits are append-only in the audit history. Export includes the rubric version and reviewer rationale.

**What it shows:** Keeps a time-ordered record of the AI suggestion, human decision, reviewer, rationale, evidence, and rubric version. **Why it matters:** A durable decision trail makes later review, correction, and export more transparent without erasing how a conclusion was reached.

# Rubric versioning with a transparent change log

Assessment configuration · Keeps past results tied to the rubric version used at the time.

Rakamin AI Interview | Assessments | Vacancies | Review | Tenant: Demo Hiring Team | Help

Vacancies / Senior Frontend Engineer / Rubric versions

## Rubric versions

Published versions are immutable. New wording applies only to future sessions.

Create draft version

### Version history

| VERSION | STATUS | PUBLISHED | USED BY |
|---|---|---|---|
| **v2.1** Current · Frontend Engineer | Published | 12 Oct 2026 | 8 sessions |
| v2.0 | Archived | 04 Sep 2026 | 19 sessions |
| v2.2 draft | Draft | Not published | 0 sessions |

Existing portfolios keep their original rubric snapshot.

### Draft change preview · v2.2

2 changes

REACT ARCHITECTURE · LEVEL 3

~~Explains common trade-offs.~~

**Now:** Explains a relevant trade-off and a validation approach, with an example.

TESTING STRATEGY · LEVEL 2

Added an explicit unit/integration boundary example.

Compare full rubric | Request review

**What it shows:** Shows published and draft rubric versions, a change preview, and which sessions used each immutable version. **Why it matters:** Versioning keeps historical results interpretable and prevents a rubric edit from silently changing the standard used for future or past candidates.

# Assessor calibration with shared practice cases

Team calibration · Uses synthetic examples and compares reasoning, not just numbers.

Rakamin AI Interview | Assessments | Vacancies | Review | Tenant: Demo Hiring Team | Help

Team tools / Calibration / Frontend rubric v2.1

## Calibration exercise

Practice case 04 of 06

Practice together using fictional candidate evidence. No live candidate records are included.

### Fictional response · React architecture

Synthetic

> "I used a shared store for the filters because the results panel and the filter drawer both needed the selected values. I'd add a test around the reset behavior; we had a bug where it cleared the URL but not the input."

RUBRIC ANCHOR

**Level 2:** Describes a workable approach with limited trade-off analysis.

**Level 3:** Explains a suitable structure, trade-offs, and common failure modes.

What evidence supports your rating? Add rationale

### Your rating

L2 | L3 | L4 | Insufficient

The example identifies shared state and a failure mode...

### Team comparison

You · Level 3 — 5 reviewers submitted

Rationale discussion opens after everyone submits. Names are hidden during the first pass.

Submit practice rating

**What it shows:** Lets assessors rate fictional examples independently, explain their reasoning, and compare interpretations after submission. **Why it matters:** Calibration can expose ambiguous rubric wording and improve consistency without using live candidate data as practice material.

# Aggregate fairness monitoring with privacy safeguards

Quality oversight · Highlights patterns for qualified review; it does not infer sensitive traits.

Rakamin AI Interview | Assessments | Vacancies | Review | Tenant: Demo Hiring Team | Help

Team tools / Assessment quality / Aggregate patterns

## Assessment consistency

Last 90 days · aggregate only

Use patterns to decide what to investigate—not to label individual candidates.

**REVIEWED SESSIONS**
126
Across 4 role rubrics

**HUMAN REVIEW COVERAGE**
94%
Portfolios with recorded rationale

**OPEN QUALITY CHECKS**
2
Require qualified reviewer

### Rating distribution by interview language

| LANGUAGE | SESSIONS | EVIDENCE COVERAGE | REVIEW NOTE |
|---|---|---|---|
| English | 84 | 82% | No alert |
| Bahasa Indonesia | 42 | 76% | Check prompt parity |

No protected traits are inferred. Small groups are hidden below the minimum reporting threshold.

### Review queue

**Prompt parity check** — Needs review

Compare translation and probe coverage for rubric v2.1. Aggregate signal only; not a finding of bias.

Assign qualified reviewer

This view supports governance. It must not be used to auto-rank or reject candidates.

**What it shows:** Displays aggregate process signals and a qualified-review queue while suppressing small groups and avoiding individual rankings. **Why it matters:** Teams need a cautious way to investigate possible process differences; aggregate signals should prompt human inquiry, not be treated as proof or used to decide an individual's outcome.

# Data inventory and retention policy controls



Rakamin AI Interview | Assessments | Vacancies | Review | Tenant: Demo Hiring Team | Help

Admin / Privacy controls / Data inventory

## Data inventory & retention

Draft · review required

Draft controls for legal and operations review before any production policy is published.

### Data categories

| RECORD | PURPOSE | ACCESS | RETENTION |
|---|---|---|---|
| Audio stream | Live interview only | Candidate session service | Do not persist* |
| Transcript | Evidence review | Assigned hiring team | Employer policy |
| Portfolio & decisions | Role assessment record | Authorized reviewers | Employer policy |
| Exports | Hiring workflow | Explicitly authorized | Short-lived link |

* Proposed default shown for discussion; confirm actual system behavior before publishing.

### Policy owners

| | |
|---|---|
| Product / Operations | Assigned |
| Security reviewer | Pending |
| UU PDP counsel | Required review |

### Deletion workflow

Define request validation, authorization, deletion scope, backups, and completion evidence.

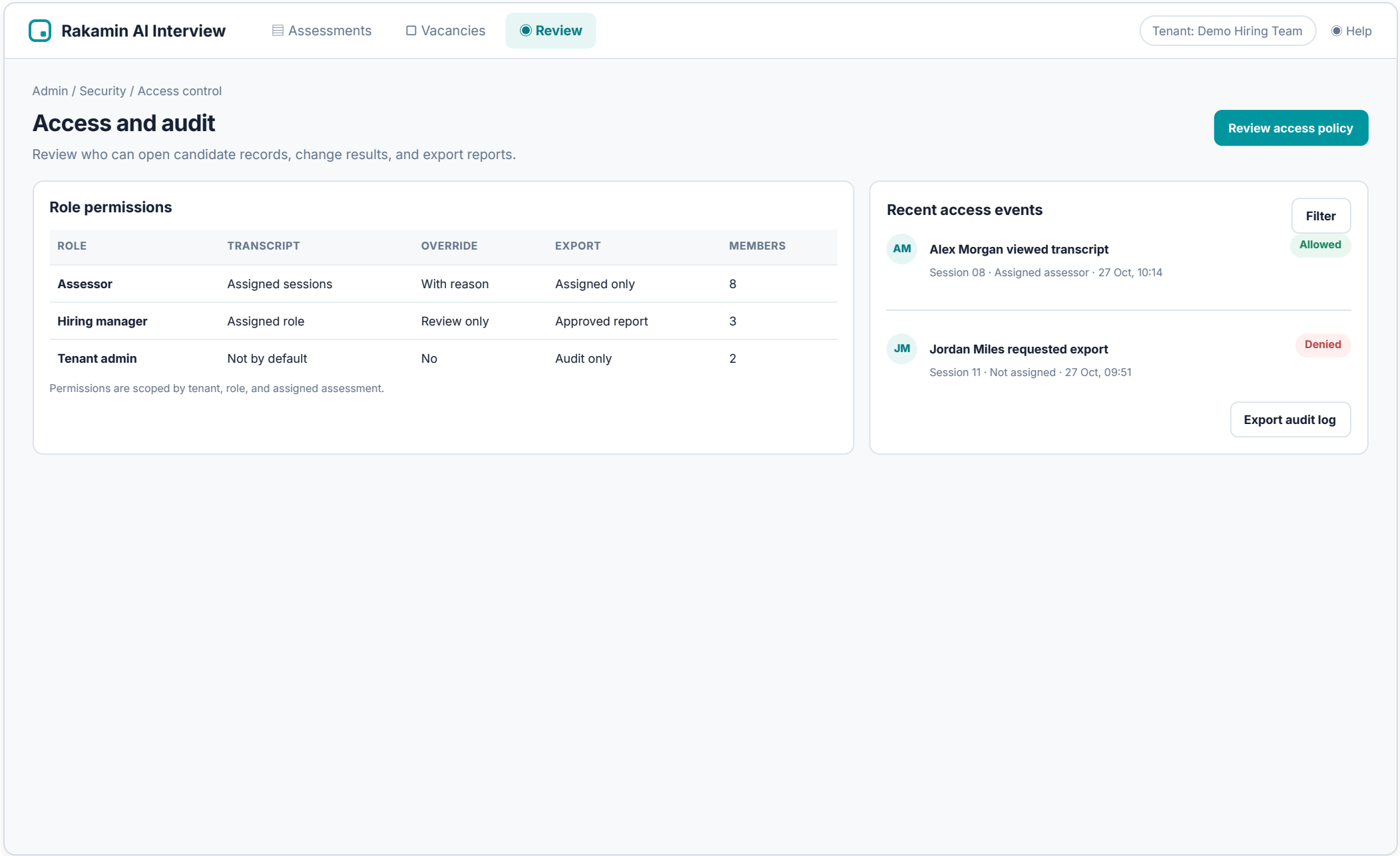
Configure request process

No policy changes take effect until approved and versioned.

**What it shows:** Lists data categories, purpose, access, proposed retention, deletion workflow, and the people responsible for policy review. **Why it matters:** Explicit ownership and lifecycle rules reduce accidental over-retention and help teams obtain security and legal review before publishing policy.

# Role-based access and auditable record access

Operations · Gives administrators a least-privilege view of permissions and access events.




**What it shows:** Summarizes role permissions and records who accessed a candidate record, including denied attempts. **Why it matters:** Least-privilege access and an audit trail help protect sensitive records and support investigation when access is questioned.

# Queue health and recoverable provider failures

Operations · Shows actionable job states and categorized errors without exposing secrets.

Rakamin AI Interview | Assessments | Vacancies | Review | Tenant: Demo Hiring Team | Help

Admin / Operations / Queue health

## Background jobs & provider status

Operational signals are separate from candidate-facing results.

Refresh · 12:34:08

**QUEUE**

4

Healthy

1 running · 3 waiting

**PROVIDER**

Gemini

Rate limited

Retry after 42 seconds · no credentials shown

**LAST SUCCESSFUL ANALYSIS**

11:58

Synthetic test workspace

### Recent jobs

| JOB | STATE | ATTEMPT | ACTION |
|---|---|---|---|
| Portfolio analysis · #8401 | Retry scheduled | 2 / 4 | View safe error |
| Fit-gap report · #8398 | Complete | 1 / 4 | View result |
| Portfolio analysis · #8392 | Needs attention | 4 / 4 | Inspect |

### Failure detail · #8401

Quota / retryable

**SAFE DIAGNOSTIC**

Provider request throttled. Retry scheduled with backoff. Candidate report remains “processing”—no empty success result.

Idempotency key verified · duplicate write prevented

Open trace

Trace view redacts tokens, prompt payloads, and candidate content.

**What it shows:** Shows queue and provider health, categorized retryable failures, safe diagnostics, and whether duplicate writes were prevented. **Why it matters:** Operators can recover jobs without mistaking failure for success, exposing candidate information, or creating duplicate results.

# Candidate comparison against role requirements







## Role-aligned candidate comparison

Rubric v2.1 · 3 reviewed sessions

Compare evidence against the same published rubric. No unsupported “best candidate” score is calculated.

Each candidate has a different evidence profile. Use the role requirements and human review notes; this table does not recommend a hiring decision.

| ROLE CRITERION | JORDAN LEE | SAMIRA PUTRI | CHRIS TAN |
|---|---|---|---|
| **React architecture**<br>Required · rubric v2.1 | Level 3 · reviewed<br>2 evidence excerpts<br>Open evidence | Level 3 · reviewed<br>3 evidence excerpts<br>Open evidence | Insufficient<br>1 of 3 probes<br>Follow up needed |
| **Testing strategy**<br>Required · rubric v2.1 | Level 4 · reviewed<br>Human rationale available | Level 2 · reviewed<br>Human rationale available | Level 3 · reviewed<br>Human rationale available |
| **Accessibility**<br>Required · rubric v2.1 | Follow-up suggested | Level 3 · reviewed | Level 3 · reviewed |

Select criteria · Export accessible comparison

Names and interview records are visible only to authorized reviewers. Criteria must be job-related and approved.

**What it shows:** Compares reviewed evidence across candidates against the same approved role criteria, with gaps clearly marked and no aggregate rank. **Why it matters:** Hiring teams can discuss job-relevant evidence while avoiding an unsupported composite score that hides different evidence quality.

# Shared assessment templates with ownership and versions



Rakamin AI Interview | Assessments | Vacancies | Review | Tenant: Demo Hiring Team | Help

Team tools / Assessment templates

## Assessment templates

Share approved role setups with named owners, version history, and clear use permissions.

Create template

### Team library

Search templates

| TEMPLATE | OWNER | VERSION | ACCESS |
|---|---|---|---|
| **Frontend Engineer**<br>45 min · English · 5 skills | Alex Morgan | v2.1 published | Hiring team |
| **Backend Engineer**<br>60 min · English · 6 skills | Priya Shah | v1.3 review | Engineering |
| **Product Designer**<br>45 min · Bahasa / English | Rina S. | v1.0 draft | Private |

### Frontend Engineer · v2.1

Published

Owner: Alex Morgan · Updated 12 Oct 2026

INCLUDED SKILLS

- React architecture — Required
- Testing strategy — Required
- Accessibility — Required

5 authorized editors

Manage access

View versions

Use as starting point

**What it shows:** Provides a shared template library with named owners, version status, included skills, and controlled access. **Why it matters:** Reusable setups reduce repetitive work, while ownership and versioning prevent accidental changes to an assessment already in use.

# Evidence-linked reviewer discussion and accessible export

Hiring workflow · Keeps comments attached to evidence and labels AI versus human conclusions.

Rakamin AI Interview | Assessments | Vacancies | Review | Tenant: Demo Hiring Team | Help

Assessments / Senior Frontend Engineer / Session 08 / Review

## Team review & report

Rubric v2.1 · Human-reviewed

Discuss evidence, assign follow-ups, and export a versioned, accessible record.

### React architecture · Level 3

Human decision

> “I split the search state from the results list so typing doesn’t trigger a full page refresh.”
> Transcript evidence · 18:42

PS

**Priya Shah · Hiring manager**

Can we ask for a concrete example of how this was tested?

Follow-up assigned to assessor

Add a comment linked to this evidence...

### Follow-up questions

☐ Ask candidate about validation approach

Add question

### Export report

Includes evidence excerpts, rubric version, confidence, assessor rationale, and review history.

| | |
|---|---|
| AI suggestion | Clearly labeled |
| Human decision | Rationale included |
| Accessibility | Tagged PDF |

EXPORT FORMAT

Accessible PDF | JSON

Only authorized reviewers can export. The file records the rubric version and export time.

Preview report

**What it shows:** Connects reviewer comments and follow-up questions to evidence, then offers a rubric-versioned, accessible export that labels AI and human conclusions. **Why it matters:** Discussion stays grounded in the underlying response, and recipients can understand the source and review history of each conclusion.